

ComGround: Measuring Common Ground in LLMs through Category Norms and Tacit Coordination

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We present **ComGround**, a benchmark for evaluating whether large language models (LLMs) share the *common ground* that underpins human social cognition—the shared knowledge, beliefs, and cultural salience that enable communication and coordination. ComGround comprises two complementary tasks. **Task 1** (Category-Norm Replication) directly replicates a classic cognitive psychology paradigm: given a category label, the model freely generates exemplars, and the resulting frequency distribution is compared against human normative data from US participants across 70 categories. **Task 2** (Tacit Coordination) tests whether this knowledge supports implicit coordination: two independent agents must converge on the same option from a shuffled 4-option set without communication, relying solely on shared knowledge across 410 items in 31 domains. Evaluating six frontier models, we find that while all models capture the ordinal structure of human category norms ($\rho > 0.74$), fine-grained alignment remains weak (first-response match < 0.56 , Pearson $r < 0.35$). In tacit coordination, models achieve 0.53–0.88 success rates (chance = 0.25), with substantial variation across knowledge domains and a persistent gap between culture-specific and universal knowledge. These results reveal both the promise and limitations of LLMs as agents capable of the shared understanding required for trustworthy human–AI collaboration and multi-agent coordination.

CCS Concepts: • **Computing methodologies** → **Natural language processing**.

Additional Key Words and Phrases: social cognition, category norms, tacit coordination, focal points, common ground, large language models

ACM Reference Format:

Xinyue Peng. 2026. ComGround: Measuring Common Ground in LLMs through Category Norms and Tacit Coordination. 1, 1 (May 2026), 16 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

A core component of social cognition is *common ground*—the shared knowledge, beliefs, and assumptions that people expect others to hold [6]. When asked to name “a fruit,” most English speakers say “apple” first; when asked to independently name a US city for a coordination bonus, most say “New York.” Both phenomena reveal the same underlying capacity: shared cultural knowledge that makes certain items more *salient* than others. Measuring whether LLMs possess this common ground is critical for at least three reasons: (1) AI agents that share human salience structure can serve as more reliable collaborators in high-stakes settings (e.g., emergency response, medical decision-making); (2) multi-agent systems require shared expectations to coordinate without explicit communication; and (3) alignment between human and model knowledge structures is a prerequisite for trustworthy AI that behaves predictably in social contexts.

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ACM XXXX-XXXX/2026/5-ART

<https://doi.org/XXXXXXX.XXXXXXX>

We propose **ComGround**, a benchmark comprising two tasks that probe this capacity at different levels:

Task 1 (Category-Norm Replication) directly replicates a well-established cognitive psychology paradigm. We adopt the 70-category semantic generation task from Castro, Curley, & Hertzog [5], in which participants listed as many exemplars as possible for each category label. We transfer this task to LLMs with minimal modification and compare model-generated exemplar distributions against the original human norms. This tests whether LLMs have internalized the same category structure as humans—not just which items belong to a category, but how *accessible* each item is.

Task 2 (Tacit Coordination) goes further: it tests whether shared salience knowledge can support *coordination* between two independent agents. Drawing on Schelling’s [17] focal-point theory, we present two LLM agents with the same 4-option category but with options shuffled in different orders, and ask them to independently choose the same option. Success requires both agents to converge on the same culturally salient item without communication.

We organize our investigation around four research questions:

- **RQ1:** Do LLMs internalize the same category-norm structure as humans?
- **RQ2:** Can independent LLM agents achieve tacit coordination through shared salience?
- **RQ3:** How do knowledge domain and cultural context affect coordination?
- **RQ4:** Does category-norm alignment predict tacit coordination success?

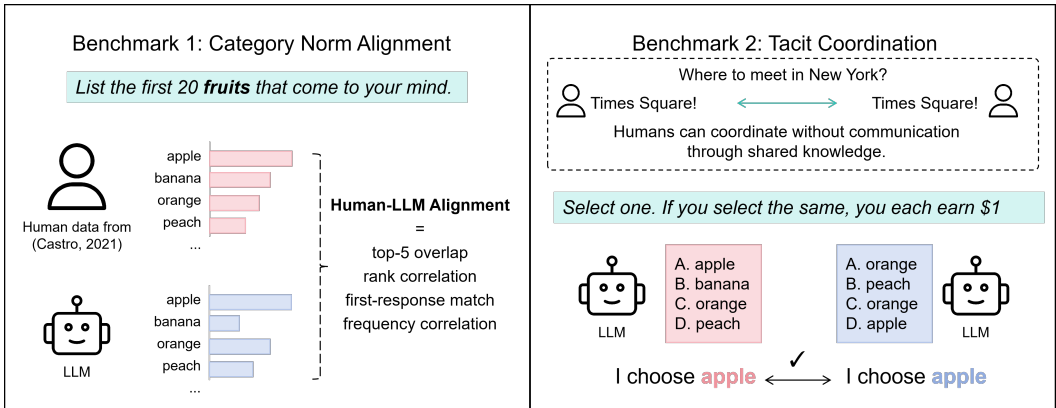


Fig. 1. Conceptual overview. Left: category norms reveal the accessibility structure of semantic memory. Right: two independent agents converge on the same option without communication, relying on shared salience.

2 Related Work

Category norms and semantic knowledge. When humans generate exemplars for a semantic category, the resulting frequency distribution reveals the *accessibility* structure of semantic memory—some items are retrieved far more readily than others. This finding was first systematically documented by Battig & Montague [3], who collected exemplar norms for 56 categories from over 400 participants, revealing robust typicality gradients across domains. Van Overschelde, Rawson, & Dunlosky [22] updated and expanded these norms to 70 categories, and Castro, Curley, & Hertzog [5] provided the most recent revision with both frequency and age-of-acquisition norms. These category norms have become foundational tools in cognitive psychology, used to study semantic

memory organization [5], typicality effects, and cultural knowledge structure. The key insight is that *shared* exemplar frequency rankings reflect shared cultural knowledge: when most English speakers say “apple” first for “a fruit,” this consensus reveals common ground about category structure.

Focal points and tacit coordination. Schelling’s [17] seminal work demonstrated that people can coordinate without communication by recognizing *focal points*—outcomes that “stand out” due to salience, convention, or shared knowledge. In pure coordination games, where players share identical interests but cannot communicate, the ability to converge on the same option depends entirely on shared expectations about what is salient. Mehta, Starmer, & Sugden [16] provided decisive experimental evidence: when asked to independently name the same item from a set, participants achieved coordination rates far above chance, driven by cultural salience rather than strategic reasoning. Sugden [20] formalized this as a theory of focal points, and Crawford & Haller [9] showed how agents learn to coordinate through repeated interaction. This literature connects directly to Clark’s [6] theory of common ground in language use: successful communication and coordination both rest on the assumption that interlocutors share knowledge about what is culturally salient [7, 8]. Lewis [15] further showed that conventions—including coordination equilibria—can arise from shared expectations alone.

LLM social cognition and theory of mind. A growing body of work examines whether LLMs possess social cognition capacities analogous to humans. Kosinski [14] argued that theory of mind (ToM) may have spontaneously emerged in LLMs, as GPT-4 performed well on classic false-belief tasks. However, Strachan et al. [19] conducted a large-scale, preregistered study showing that LLMs’ ToM performance is fragile and sensitive to minor prompt variations, while Ullman [21] demonstrated that trivial alterations to ToM tasks cause LLMs to fail. These findings suggest that current LLMs may be solving ToM tasks through pattern matching rather than genuine social understanding. In the domain of strategic interaction, Brookins & DeBacker [4] found that GPT-3 plays economic games (dictator, ultimatum, trust) with reasonable strategies, and Akata et al. [2] showed that LLMs can learn to cooperate in repeated games. Horton [12] proposed using LLMs as “Homo silicus”—simulated economic agents—while Aher et al. [1] explored using LLMs to simulate multiple human participants.

Our contribution. Despite this progress, no existing benchmark specifically tests whether LLMs *share the cultural salience structure* of human semantic knowledge. Category norm studies [3, 5, 22] have been conducted exclusively with human participants and have not been applied to measure LLM–human alignment in exemplar accessibility. Similarly, while focal-point experiments [13, 16, 17] reveal human coordination abilities, no systematic benchmark tests whether two independent LLMs can achieve tacit coordination through shared salience. Existing LLM benchmarks [11, 18, 23] measure factual knowledge and reasoning but not the *structure* of shared cultural knowledge. Our benchmark, ComGround, fills this gap: Task 1 directly measures LLM–human alignment in category-norm structure, and Task 2 tests whether that shared knowledge enables tacit coordination between independent agents. This aligns with the Kaggle Measuring Progress Toward AGI competition [10], which identifies social cognition as a critical but understudied dimension of AI capability. Our results demonstrate that common ground is measurable, varies across models and domains, and does not trivially follow from factual knowledge.

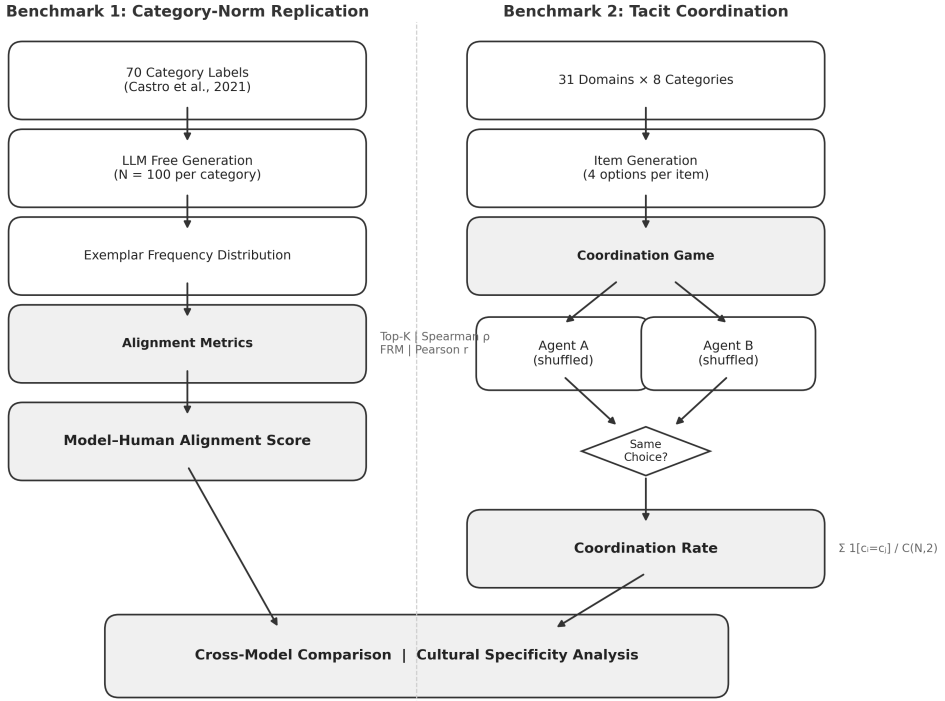


Fig. 2. ComGround pipeline. Task 1: LLM generates exemplars for 70 categories; four alignment metrics compare against human norms. Task 2: two isolated agents choose from shuffled options; pairwise agreement yields the coordination rate.

3 Task 1: Category-Norm Replication

3.1 Task

We adopt the 70 category labels from Castro, Curley, & Hertzog [5] verbatim (Table 1). For each category, the LLM is prompted:

Generate 20 exemplars for the category “[category label]”. Write one example per line.

This mirrors the original human task, where participants had 30 seconds to generate exemplars freely. We run each prompt once per category and compare the ranking of generated exemplars against human frequency norms.

3.2 Categories

The 70 categories span a broad range of semantic domains. A subset is shown below; the full list is in the supplementary materials.

3.3 Alignment Metrics

We compare LLM-generated exemplar distributions against the human normative data from [5] using four metrics:

Table 1. Task 1: Selection of the 70 category labels (Castro et al., 2021).

Domain	Sample Categories
Natural kinds	A bird; A fish; A flower; A fruit; A tree
Artifacts	A type of car; A type of fabric; A weapon
Social	A female first name; An occupation; A sport
Spatial	A city; A state; A country
Abstract	A color; A unit of time; A part of speech
Total	70 categories

(1) Top- K overlap. The Jaccard similarity between the top- K most frequent exemplars from humans and the LLM:

$$\text{TopK}(cat, K) = \frac{|H_K \cap L_K|}{|H_K \cup L_K|} \quad (1)$$

where H_K and L_K are the sets of top- K exemplars for humans and LLM respectively.

(2) Rank correlation. Spearman’s ρ between human and LLM frequency rankings across exemplars appearing in either top-20 list.

(3) First-response match. Whether the LLM’s most frequent exemplar matches the human modal response:

$$\text{FRM}(cat) = \#[\arg \max_L = \arg \max_H] \quad (2)$$

(4) Frequency correlation. Pearson’s r between human and LLM frequency vectors over shared exemplars.

The overall Task 1 score is the average across all 70 categories.

3.4 Implementation

We implement the benchmark using the Kaggle Benchmarks (kbench) framework. A sub-task (generate_exemplars) prompts the LLM with each category label and returns the raw response. The main task (category_norm_alignment) evaluates this sub-task over all 70 categories, parses exemplars from the responses, and computes alignment against human norms using the four metrics above. The composite score is the average of (Top-5 overlap, $\max(\rho, 0)$, first-response match, $\max(r, 0)$) across categories. The overall score is reported as tuple(float, float): (mean alignment, standard deviation).

4 Task 2: Tacit Coordination

4.1 Task

Two independent LLM agents play the following coordination game:

You are playing a coordination game with another agent. If you both select the SAME option, you each earn \$1. You cannot communicate. Note: The other agent sees the SAME options, but in a DIFFERENT order.

Each agent is shown a neutral category label and four options in a *randomized* order. Option order is shuffled independently for each agent, preventing trivial coordination via position bias.

4.2 Design Principles

(1) No leading language. Categories are strictly neutral—no “most famous,” “traditional,” “best,” or “typical.” Coordination depends solely on the agent’s own salience judgment.

(2) Four options at the same semantic level. All four options are roughly equally plausible, no single option being clearly more salient than the others.

Table 2. Task 2: Domain taxonomy. 31 domains in 8 categories.

Category	Domains (<i>n</i>)	Primary Cognitive Mechanism	Culture
Perception	Colors; Shapes; Spatial Directions; Extremes (4)	Perceptual prototype; symmetry bias; spatial default	Unspecific
Symbolism	Numbers; Time Anchors; Emotions (3)	Symbolic salience; roundness; temporal convention	Unspecific
Biology	Animals; Plants; Fruits; Body Parts; Senses (5)	Biological prototype; familiarity; embodied centrality	Unspecific
Artifacts	Tools; Clothing; Vehicles; Furniture (4)	Functional typicality; affordance salience	Unspecific
Places	Rooms; Public Places; Institutions; Geographic Entities (4)	Spatial routine; meeting-point scripts; collective prominence	Mixed [*]
Norms	Family Roles; Occupations; Social Norms (3)	Role prototype; prestige salience; normative scripts	Mixed [*]
Culture	Holidays; Food; Drinks; Famous People; Media; Brands (6)	Collective memory; ritual salience; media exposure	Specific
Digital	Digital Platforms; Internet Culture (2)	Platform familiarity; interface conventions	Specific
Total	31 domains		

^{*}Mixed: most domains unspecific; Public Places, Geographic Entities, Social Norms are culture-specific.

(3) Bilingual and cross-cultural. Every item exists in both English and Chinese. Culture-sensitive domains include separate US-context and China-context items.

4.3 Domain Taxonomy

We organize 31 domains into 8 macro-categories based on the primary *cognitive mechanism* through which humans are hypothesized to achieve coordination in each domain, drawing on prior research in cognitive psychology and focal-point theory (Table 2). For instance, coordination on colors or shapes likely relies on perceptual prototypes, while coordination on holidays or brands depends on collective cultural memory.

These mechanisms differ in whether successful coordination requires culture-specific knowledge, which we use to divide the 8 macro-categories into three groups. **Culture-unspecific** categories (Perception, Symbolism, Biology, Artifacts) draw on mechanisms broadly shared across communities, such as perceptual salience, symbolic convention, and functional typicality. **Culture-specific** categories (Culture, Digital) depend on conventions that vary substantially across communities. **Mixed** categories (Places, Norms) contain both, with three domains (Public Places, Geographic Entities, Social Norms) relying on culture-specific knowledge.

This distinction matters for LLM evaluation because culture-specific focal points are acquired through exposure to particular communities’ texts and conventions. Models trained on different corpora may thus internalize different salience structures, leading to coordination failures not from general reasoning deficits but from mismatched cultural representations. Varying cultural specificity across domains allows ComGround to diagnose where and why LLM coordination breaks down.

Each culture-unspecific domain contributes 10 items shared across English and Chinese conditions, totaling 210 items across 21 domains. Each culture-specific domain contributes 10 US-context items in English and 10 China-context items in Chinese, totaling 200 items across 10 domains. The full benchmark comprises **410 items** (205 English, 205 Chinese).

4.4 Evaluation: Coordination Rate

For each item, we query the model 100 times with randomized option orders, simulating 100 agents. The pairwise coordination rate is:

$$\text{CoordRate}(\text{item}) = \frac{\sum_{i < j} \mathbb{I}[c_i = c_j]}{\binom{100}{2}} \quad (3)$$

Chance is 0.25 (random among 4); perfect coordination is 1.0.

4.5 Implementation

The coordination game is implemented using kbench’s multi-agent chat isolation. Two independent agent contexts are created via `kbench.chats.new()`, each receiving the same category but with independently shuffled option orders. Responses are parsed using a 5-strategy matching algorithm (exact match, punctuation-stripped match, cue-phrase extraction, longest substring match, quoted/bolded text extraction) to robustly identify the chosen option. The sub-task returns a boolean (coordinated / not coordinated), and the main task aggregates over all items to return (coordination rate, standard deviation).

4.6 Item Quality Analysis

Before running LLM evaluation, we assess the quality of the 410 items along two dimensions: *semantic distance* and *corpus frequency*. A well-designed item should have four options that are indistinguishable on these surface-level metrics: if one option is substantially more frequent or semantically distinct than the others, models could coordinate simply by selecting the statistical outlier, without drawing on any internalized knowledge. We therefore treat low salience gaps on both dimensions as a quality criterion, and verify that most items in the benchmark successfully resist this kind of surface-level shortcut.

Semantic distance. For each item, we embed all four options using `text-embedding-3-large` and compute pairwise cosine distances. Each option receives an *average distance* score—the mean cosine distance to the other three options. Higher values indicate the option is more semantically distinct from the alternatives.

Corpus frequency. We use the `wordfreq` library, which aggregates word frequencies from Google Books, Wikipedia, and SUBTLEX subtitle corpora, to obtain Zipf-scale log frequencies for each option in its respective language (English or Chinese). Higher Zipf values indicate more common words in the corpus.

Results. Across all 1,640 options (410 items $\times$ 4 options), the mean average semantic distance is 0.54 ($\sigma = 0.10$), and the mean corpus frequency is 3.89 on the Zipf scale ($\sigma = 0.97$).

To assess *focal clarity*—whether each item has a salient option that “stands out”—we compute the *gap* between the top-1 and top-2 options for both metrics (Appendix, Figure 8). The mean distance gap is 0.035 ($\sigma = 0.030$) and the mean frequency gap is 0.40 ($\sigma = 0.42$), indicating that corpus frequency provides stronger focal differentiation than embedding distance. Per-domain analysis reveals that Digital and Culture domains show the largest frequency gaps, while perceptual domains (Colors, Shapes) show the smallest.

We further quantify focal clarity using Shannon entropy of the normalized 4-option distributions (Figure 3). Distance entropy is uniformly high (mean = 1.997 bits, max = 2.0), confirming that embedding distances are nearly uniform across options within most items. Frequency entropy shows more variation (mean = 1.978, $\sigma = 0.045$), with some items showing clear frequency dominance. The English–Chinese comparison shows no significant difference in distance-based metrics ($p = 0.29$),

but English items have significantly lower frequency entropy ($p < 0.001$). Finally, the option-level scatter confirms that semantic distance and corpus frequency are weakly correlated ($r = -0.25$), capturing different aspects of salience (see Appendix A).

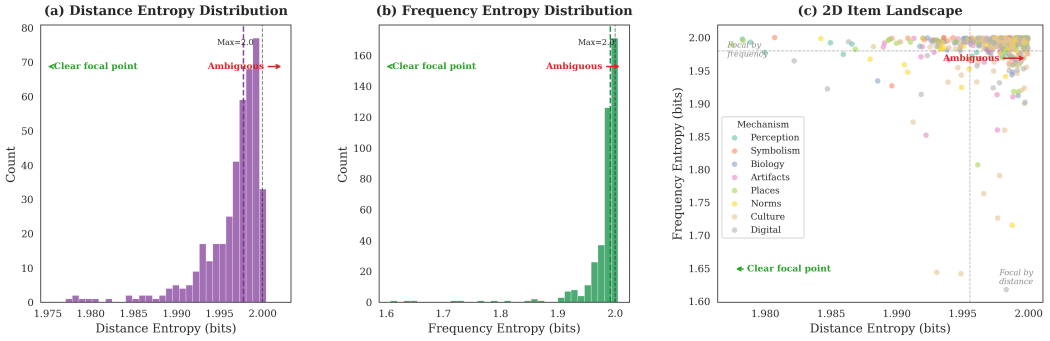


Fig. 3. Shannon entropy of within-item option distributions. Most items cluster near maximum entropy (2.0 bits), meaning the four options are nearly indistinguishable by embedding distance or corpus frequency alone. This confirms that successful coordination cannot rely on surface-level statistical cues but requires models to leverage internalized human knowledge.

5 Results

5.1 Experimental Setup

We evaluate six frontier models—Gemini 2.5-Flash, Gemma 4, GLM-5, Claude Sonnet 4.6, DeepSeek-R1, and GPT-oss—using the Kaggle Benchmarks framework. For Task 1, each category is prompted once and the model freely generates exemplars; alignment with human norms is computed using the four metrics in Section 3.3. For Task 2, each item is run with 100 simulated agents (randomized option orders per agent); pairwise agreement yields the coordination rate.

5.2 RQ1: Do LLMs Internalize Human Category-Norm Structure?

Table 3 and Figure 4 present the alignment results.

All models achieve strong rank correlation with human norms ($\rho > 0.74$), confirming that LLMs broadly capture the ordinal structure of human exemplar frequency (Figure 4c). However, the picture diverges noticeably across metrics. First-response match (FRM) ranges from 0.314 (GPT) to 0.557 (Gemini), and top-5 overlap from 0.314 to 0.456, indicating that the *fine-grained* accessibility structure—which specific exemplars are most readily retrieved—remains substantially misaligned. Pearson frequency correlation is uniformly low ($r < 0.35$), suggesting that while models rank exemplars similarly to humans, they do not reproduce the relative frequency magnitudes. Coverage reveals a further gap: models recover only 552–766 of the 1,561 distinct exemplars produced by human participants (35–49%), indicating that a substantial portion of human semantic memory remains outside the models’ generative vocabulary for these categories.

Figure 5 breaks down coverage by semantic domain. All models show higher coverage for Natural Kinds and Abstract categories and lower coverage for Social and Spatial categories. This pattern suggests that LLMs may have more robust representations for categories grounded in perceptual and definitional features (e.g., animals, colors) than for those relying on socially and culturally contingent knowledge (e.g., occupations, places).

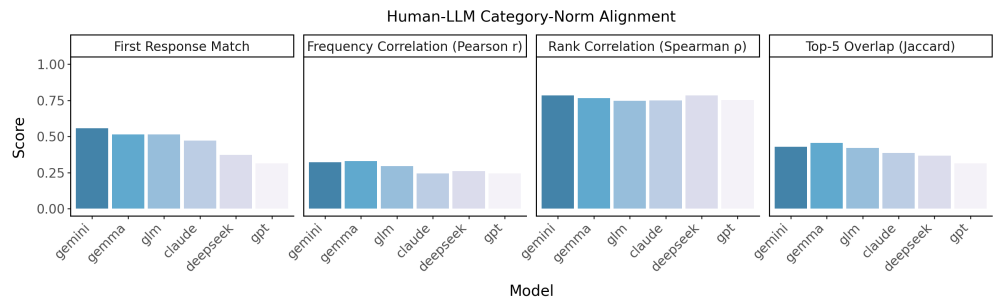


Fig. 4. Task 1 results by model and metric. All models achieve strong rank correlation ($\rho > 0.74$) but diverge on first-response match and frequency correlation, revealing a gap between ordinal and magnitude alignment with human norms.

Table 3. Task 1 results: alignment with human category norms (70 categories). Composite = mean of the four metrics. Coverage = distinct human exemplars generated (out of 1,561). Full names: Gemini = google/gemini-2.5-flash; Gemma = google/gemma-4-26b-a4b; GLM = zai/glm-5; Claude = anthropic/claude-sonnet-4-6; DeepSeek = deepseek-ai/deepseek-r1-0528; GPT = openai/gpt-oss-20b.

Model	Composite	Top-5	ρ	FRM	r	Coverage
Gemini	0.526	0.428	0.784	0.557	0.321	700
Gemma	0.518	0.456	0.765	0.514	0.330	766
GLM	0.497	0.421	0.747	0.514	0.295	750
Claude	0.469	0.385	0.749	0.471	0.244	687
DeepSeek	0.451	0.368	0.783	0.371	0.260	624
GPT	0.413	0.314	0.752	0.314	0.245	552

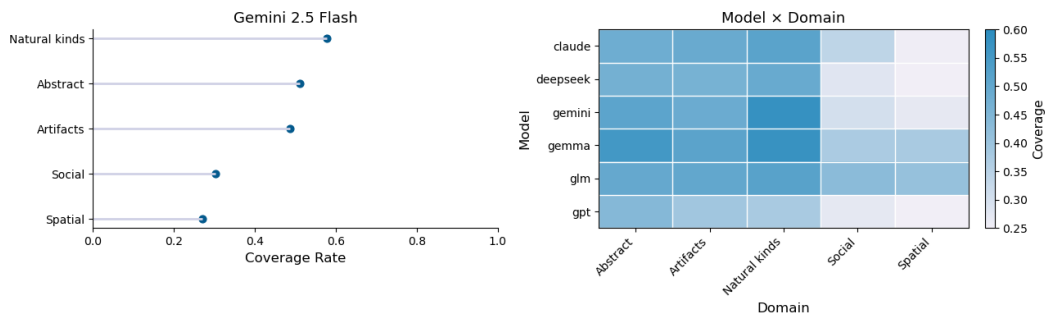


Fig. 5. Task 1 coverage by semantic domain. Higher coverage for Natural Kinds and Abstract categories; lower for Social and Spatial categories, suggesting training-data gaps in culturally specific domains.

5.3 RQ2: Can Independent LLM Agents Achieve Tacit Coordination?

Figure 6 and Table 4 present the coordination results. All six models achieve coordination rates far above chance (0.25), with DeepSeek leading at 0.880 and GPT trailing at 0.529. The top four models (DeepSeek 0.880, Claude 0.859, GLM 0.771, Gemma 0.734) demonstrate that current LLMs can achieve substantial tacit coordination through shared salience alone. Striking variation exists

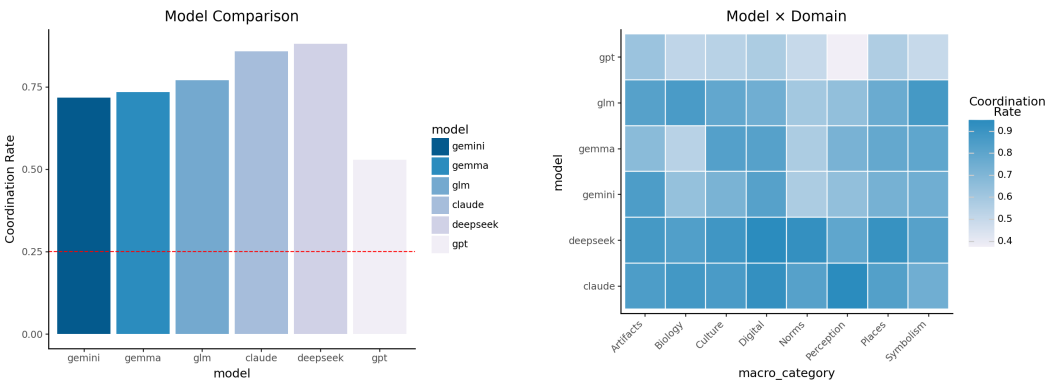


Fig. 6. Task 2 results overview. Left: overall coordination rate by model. All models far exceed chance (0.25); DeepSeek leads (0.880) and GPT trails (0.529). Right: model $\times$ domain category heatmap of coordination rates, revealing domain-specific strengths and weaknesses across models.

Table 4. Task 2 coordination rate by domain category (chance = 0.25). Bold = best per row. Model names as in Table 3.

Category	Domains	Claude	DeepSeek	Gemini	Gemma	GLM	GPT
Perception	4	0.950	0.800	0.650	0.725	0.650	0.375
Symbolism	3	0.750	0.825	0.750	0.800	0.875	0.500
Biology	5	0.880	0.840	0.640	0.540	0.860	0.520
Artifacts	4	0.850	0.875	0.850	0.675	0.825	0.625
Places	4	0.833	0.917	0.733	0.783	0.767	0.567
Norms	3	0.825	0.925	0.575	0.575	0.600	0.500
Culture	6	0.860	0.890	0.720	0.830	0.790	0.540
Digital	2	0.925	0.950	0.825	0.825	0.750	0.575
Overall	31	0.859	0.880	0.717	0.734	0.771	0.529

across domain categories: Perception yields the highest single-model rate (0.950 for Claude), while Norms shows the largest spread (0.500–0.925).

5.4 RQ3: How Do Domain and Culture Affect Coordination?

Figure 7 (left) reveals that coordination success is not uniform across domains. Claude and DeepSeek maintain high coordination rates across all eight categories, while GPT shows pronounced weakness in Perception (0.375). Domains such as Digital and Norms yield the highest overall rates, suggesting that strongly conventionalized knowledge provides the clearest focal points.

The culture gap (Figure 7, right) measures the difference in coordination rate between culture-specific items (holidays, brands, media) and universal items (colors, numbers, shapes). Most models show a positive culture gap, coordinating better on culture-specific domains where strong shared conventions exist. GLM is the sole exception with a slightly negative gap, suggesting more uniform cross-cultural knowledge representation.

5.5 RQ4: Does Category-Norm Alignment Predict Coordination?

Comparing Task 1 and Task 2 rankings reveals a striking dissociation (Tables 3 and 4). Gemini leads on Task 1 (composite 0.526) but ranks fifth on Task 2 coordination (0.717). Conversely, DeepSeek

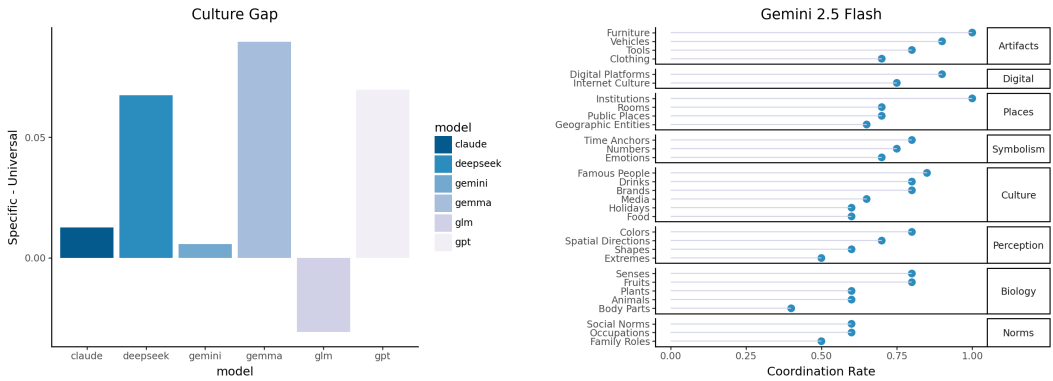


Fig. 7. Left: model $\times$ domain category heatmap. Claude and DeepSeek show consistently high coordination; GPT lags especially in Perception. Right: culture gap (culture-specific rate – universal rate). Most models coordinate better on culture-specific domains; GLM is the sole exception.

ranks fifth on Task 1 (0.451) but leads on Task 2 (0.880). Claude shows the most balanced profile (Task 1: 0.469; Task 2: 0.859). This suggests that the capacities measured by the two tasks are related but distinct: knowing *what* humans know (Task 1) does not straightforwardly translate into using that knowledge to coordinate (Task 2). The dissociation may reflect differences in how models weight cultural salience versus statistical frequency when generating versus choosing among options.

6 Discussion

Common ground as a measurable property. Our results demonstrate that common ground—the shared cultural salience structure that underpins human social cognition—can be systematically measured in LLMs. ComGround operationalizes this through two complementary tasks: Task 1 measures LLM–human alignment in the accessibility structure of semantic memory, while Task 2 measures agent–agent coordination through implicit shared knowledge. The finding that all models achieve $\rho > 0.74$ on rank correlation yet $\text{FRM} < 0.56$ reveals a nuanced picture: LLMs broadly know *what* belongs to a category but disagree with humans on *what stands out*.

The dissociation between knowledge and coordination. Perhaps the most striking finding is the dissociation between Task 1 and Task 2 rankings. Gemini leads in category-norm alignment but underperforms in coordination; DeepSeek shows the opposite pattern. This suggests that knowledge alignment and coordination ability are distinct facets of common ground. A model may accurately reflect human statistical knowledge (high ρ) yet fail to converge on the same focal point as another agent (low coordination), because coordination requires not just knowledge but a shared *decision rule* for resolving ambiguity. This has practical implications: deploying LLMs in collaborative settings (e.g., multi-agent planning, human–AI teamwork) requires measuring not just what models know, but whether multiple models will *agree* on the same action without explicit coordination.

Domain and culture effects. The strong domain dependence of coordination rates (Table 4) aligns with theories of focal-point formation: strongly conventionalized domains (Digital, Norms) provide clearer focal points than perceptually grounded ones (Perception, Biology). The culture gap analysis reveals that most models coordinate better on culture-specific items, where shared conventions create unambiguous focal points, than on universal items where multiple options are equally

plausible. GLM’s negative culture gap is noteworthy and may reflect its bilingual training data, which could attenuate culture-specific salience differences.

Practical significance. Measuring common ground in LLMs has direct applications in several areas. In *multi-agent systems*, agents that share common ground can coordinate without costly communication protocols—critical for real-time applications such as autonomous driving, disaster response, and distributed task allocation. In *AI alignment*, understanding whether models share human salience structure is a prerequisite for predictable behavior in social contexts; a model that disagrees with humans on what is most salient may make unexpected choices in high-stakes decisions. In *personalization and cultural adaptation*, the domain- and culture-specific gaps revealed by ComGround can guide targeted fine-tuning for specific user populations.

Limitations. ComGround tests a specific form of social cognition—shared cultural salience—and does not cover other aspects such as theory of mind, emotion recognition, pragmatic inference, or dynamic common-ground maintenance in dialogue. The human normative data for Task 1 comes from US participants only, limiting cross-cultural validity. Task 2 items are generated rather than empirically validated, though our item quality analysis (Section 4.6) confirms that most items are well-balanced. Additionally, our coordination game uses a single-round protocol; real-world coordination often involves repeated interaction and learning, which our benchmark does not capture. Finally, the six models evaluated represent a snapshot of current capabilities; as models evolve, the benchmark should be periodically re-evaluated.

7 Conclusion

We present ComGround, a benchmark for measuring common ground in LLMs through two complementary tasks. Task 1 (Category-Norm Replication) measures LLM–human alignment in the accessibility structure of semantic memory across 70 categories from established cognitive psychology research. Task 2 (Tacit Coordination) tests whether this knowledge enables implicit coordination between independent agents across 410 items in 31 domains spanning perceptual, cultural, and digital knowledge.

Evaluating six frontier models, we find that: (1) all models capture the ordinal structure of human category norms but fail to reproduce fine-grained accessibility; (2) coordination rates range from 0.53 to 0.88 (chance = 0.25), revealing substantial but imperfect shared salience; (3) domain and culture strongly modulate coordination success; and (4) knowledge alignment does not predict coordination ability, suggesting these are distinct facets of common ground.

Together, these tasks address a gap in current LLM evaluation: rather than testing factual knowledge or reasoning in isolation, ComGround probes whether models share the *salience structure* of human semantic knowledge—the property that makes certain items “stand out” and enables common ground between agents. This capability is essential for trustworthy human–AI collaboration and effective multi-agent coordination in real-world settings.

Acknowledgments

This work is submitted to the Kaggle Measuring Progress Toward AGI competition, Social Cognition track.

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A Additional Item Quality Figures

Figure 8 shows the focal gap analysis: the difference between the top-1 and top-2 options on distance and frequency dimensions. A larger gap implies a clearer focal point.

Figures 9 and 10 show per-item stacked distributions for semantic distance and corpus frequency, grouped by domain category.

Received 8 April 2026

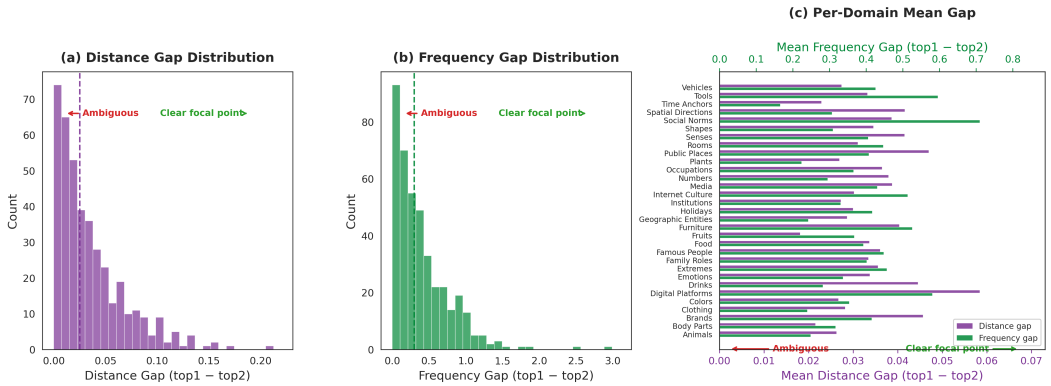


Fig. 8. Focal gap analysis. The *gap* is the difference between top-1 and top-2 options on a given saliency dimension. (a) Distance gap (mean = 0.035, σ = 0.030). (b) Frequency gap (mean = 0.40, σ = 0.42). (c) Per-domain mean gap with dual x-axes.

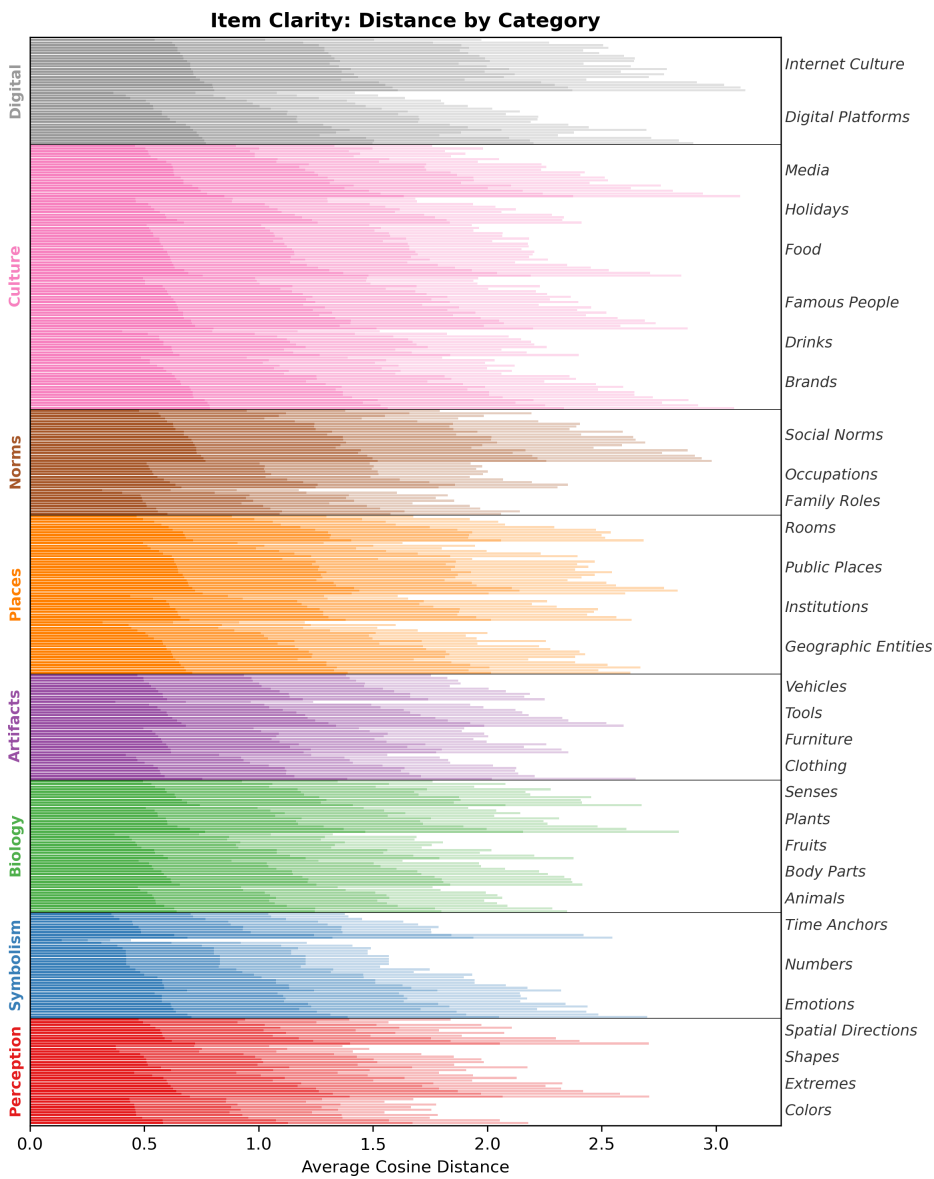


Fig. 9. Per-item embedding-distance distribution across 410 items. Stacked bars show cosine distances of four options per item, grouped by domain category.

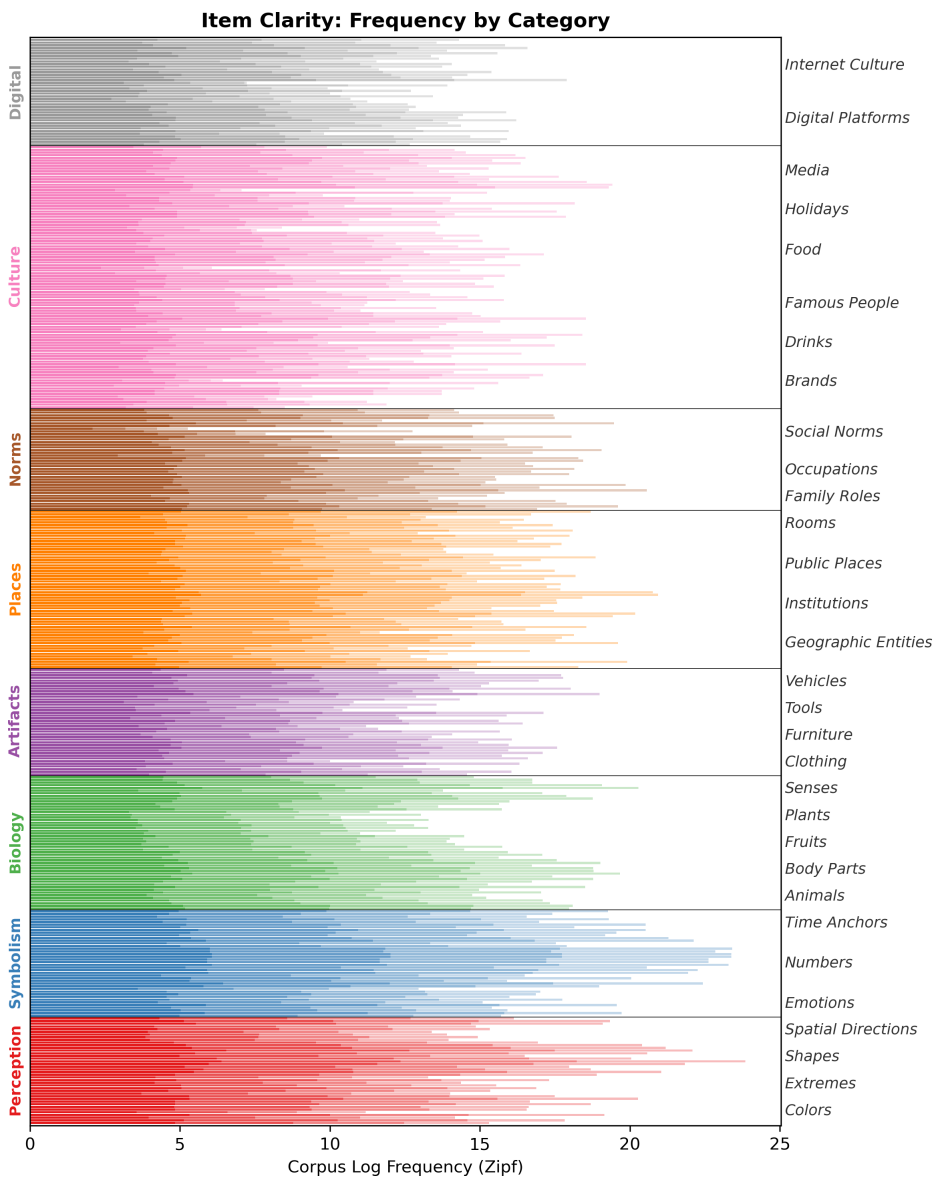


Fig. 10. Per-item corpus-frequency distribution across 405 items. Stacked bars show Zipf frequency shares; a dominant segment indicates a clear focal option.